

WASP-36b

R-0838

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_230303 · created 2026-07-28T18:40:18+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460006.6779 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.03
Transit depth (Rp/Rs)^2	1.28 +/- 0.69 [%]
Orbital Inclination (inc)	84.3 +/- 2.26
Airmass coefficient 1 (a1)	0.984 +/- 0.022
Airmass coefficient 2 (a2)	0.011 +/- 0.024
Scatter in the residuals of the lightcurve fit is	2.24 %
Best Comparison Star	#3 - [567, 324]
Optimal Aperture	3.1
Optimal Annulus	11.67
Transit Duration (day)	0.0582 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.113 $\pm$ 0.03 vs 0.1368 (deviation 0.79 σ)
Duration fit vs published	0.0582 d vs 0.0758 d (deviation 1.9 σ)
Mid-time O–C	4.1 min
Depth SNR	3.8
Residual scatter	2.24 %

Light curve

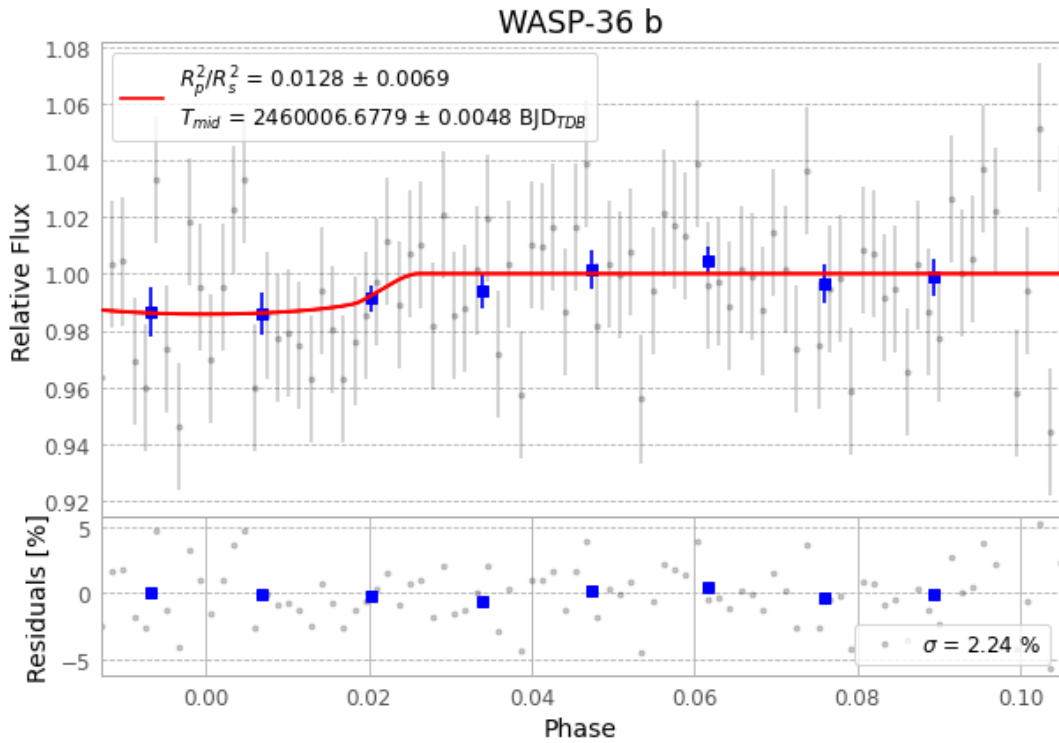


Figure 1 — FinalLightCurve_WASP-36 b_2023-03-02.png (SHA-256 3dbb81e831b0a10d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 154], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a1c6a56a98b74f7d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

88 FITS frames (58 MB), WASP-36230303033915.FITS → WASP-36230303080016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230303.log (a1760b13d2ff...), FinalLightCurve_WASP-36 b_2023-03-02.png (3dbb81e831b0...), FinalLightCurve_WASP-36 b_2023-03-02.csv (d7572cb08723...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 18:40Z.